成功大學 National Cheng Kung University 1931

穿戴科技與行動照護實驗室
Wearable Technology and Mobile Healthcare Laboratory

NEXT-GENERATION DIGITAL NEUROREHABILITATION

From Virtual to Augmented Reality Mirror Therapy *(VRMT)*

for Hemiplegic Stroke Rehabilitation

Six VR/AR mirror-therapy systems · **8 years** of NCKU research · **TFDA → FDA** path — turning *subjective rehab* into **objective neural evidence** via EEG, fNIRS & gaze.

WTMH	Wearable Technology and Mobile Healthcare
BME	Department of BioMedical Engineering
NCKU	National Cheng Kung University

PRESENTERS

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IFMBE · ARPN 2025

2026 / 06 / 16

CHAPTER 01

Foundations

The global stroke burden, classical mirror therapy, and what immersive VR adds to the box.

SLIDES 03 — 11

01

The Global Stroke Burden

Stroke remains the world's leading cause of **long-term adult disability** *— and upper-limb motor recovery is one of the largest* **unmet rehabilitation needs**.

PATTERNS OF MOTOR PARALYSIS



VRMT targets hemiplegia — one-sided motor loss seen in **~80%** of stroke survivors at onset; **~50%** remain impaired at 6 months.

BURDEN

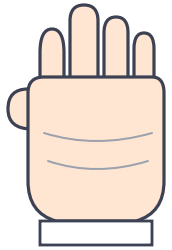
12.2M new strokes / year · 101M survivors worldwide · US$891B annual cost (≈1.1% world GDP). Scalable, evidence-based upper-limb rehabilitation is a global priority.

Stroke Hemiplegia

UNAFFECTED VS AFFECTED SIDE

Stroke causes **one-sided motor loss** *— the patient retains an intact* **unaffected (健側)** *side and an impaired* ***affected (患側)*** *side. Mirror therapy is built to exploit this asymmetry.*

UNAFFECTED SIDE · 健側



Intact motor control

- Full voluntary movement · preserved sensation & proprioception.
- Serves as the **visual driver** in the mirror loop.
- Provides the kinematic template for the cortex to re-learn.

AFFECTED SIDE · 患側



Paretic / Paralysed limb

- Weak or absent voluntary movement · spasticity, flexion synergy.
- Impaired sensation & proprioceptive feedback.
- The **rehabilitation target** — what mirror therapy aims to recover.

TARGET POPULATION

Unilateral chronic stroke patients with upper-limb hemiparesis · method: hand rehabilitation. Mirror therapy uses the unaffected side's motion to drive cortical re-organisation of the affected side.

Lawrence ES et al. Stroke 2001;32:1279–1284 · **Langhorne P et al.** Lancet 2011;377:1693–1702 · **Kwakkel G et al.** Stroke 2003;34:2181–2186.

Traditional Mirror Therapy (MT)

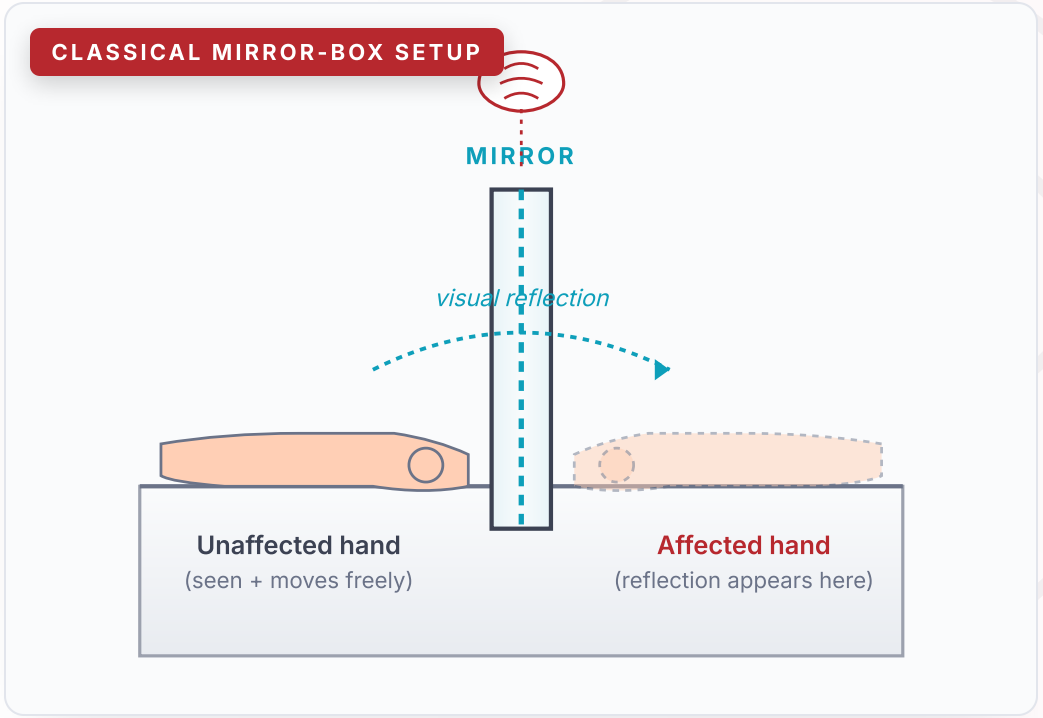
A rehabilitation method using a mirror to project the unaffected limb's motion onto the affected side — driving multisensory neuroplastic recovery.

ORIGIN & MECHANISM

- **Ramachandran & Rogers-Ramachandran (1996)** — mirror visual feedback first used for phantom-limb pain.
- **Altschuler et al. (1999)** — first application to stroke hemiparesis; brain perceives bilateral movement via the reflection.

WHY IT WORKS — NEURAL MECHANISM

- **Multisensory integration** — visual feedback substitutes for absent proprioception from the affected limb.
- **Bilateral cortical priming** — mirror feedback rebalances motor-cortex activity between hemispheres.
- **Cochrane 2018** — moderate-quality evidence for upper-limb motor recovery (62 RCTs, ~1,982 patients).



Ramachandran & Rogers-Ramachandran. Proc R Soc B 1996;263:377–386 · **Altschuler EL et al.** Lancet 1999;353:2035–2036 · **Thieme H et al.** Cochrane Database Syst Rev 2018;7:CD008449.

Traditional Mirror Therapy (MT)

CLINICAL ADVANTAGES — WHAT THE EVIDENCE SHOWS

Across 60+ RCTs and recent 2025 meta-analyses (n > 600), classical mirror therapy delivers **moderate, reproducible gains** *in four clinically meaningful domains.*

Reduces upper-limb pain

Mirror visual feedback decreases neuropathic and post-stroke shoulder/hand pain via re-mapped sensorimotor representation.

Thieme *Cochrane 2018 (n≈1,982)* · **Carrasco-Uribarren** *Healthcare 2024.*

Improves hand & grip function

Task-based mirror therapy yields measurable gains in Fugl-Meyer hand subscale, grip strength, and dexterity.

Saragih *Clin Rehabil 2025 (n=633, hand MD 1.48)* · **Arya** *Neurorehabil Neural Repair 2015.*

Reduces inter-hemispheric asymmetry

Bilateral cortical priming rebalances motor-cortex activity, reducing pathological inter-hemispheric inhibition.

Hsieh *PM&R 2025 (subacute meta)* · **Rossiter** *Neurorehabil Neural Repair 2015.*

Improves ADL functionality

Gains transfer to activities of daily living — eating, dressing, grooming — the outcomes patients care about most.

Hsieh *PM&R 2025 (ADL SMD 0.461)* · **Thieme** *Cochrane 2018.*

EVIDENCE STRENGTH

Moderate-quality evidence for upper-limb motor recovery; small but consistent effect sizes across 2018–2025 reviews. Mirror therapy is established — the open question is *how to deliver it better.*

Traditional Mirror Therapy (MT)

LIMITATIONS — THE CASE FOR IMMERSIVE VR

*The classical mirror-box constrains **what** can be trained, **where** it can be trained, and **how** sessions are dosed — gaps that an immersive VR delivery directly addresses.*

LIMITATION 1

Unable for major-action rehabilitation

The mirror box reflects only a single sagittal-plane image of simple hand poses — no shoulder, no compound multi-joint reaching.

Futuretech 2020 · Thieme Cochrane 2018.

LIMITATION 2

High environmental distraction

Open-room setup; visual and auditory noise repeatedly breaks the illusion of bilateral movement the therapy depends on.

Yıldırım, Ağrı — J Turkish Soc Algology 2016.

LIMITATION 3

Restricted to simple, ungraded activities

No task variety, no progression, no objective dose tracking — the therapist must time and grade sessions manually.

therapydna314 2020 · Thieme Cochrane 2018.

IMMERSIVE VR FIX

Full 3-D scene, complex tasks

Stereoscopic depth + ego-centric viewpoint support compound multi-joint reaching, grasping, and bimanual coordination.

IMMERSIVE VR FIX

HMD blocks the room

Head-mounted display isolates the patient from environmental noise — sustained embodiment of the affected limb.

IMMERSIVE VR FIX

Gamified tasks, auto-dose tracking

Built-in hand tracking logs joint kinematics per session; gamified tasks scale difficulty automatically.

VRMT — Virtual Reality Mirror Therapy

VRMT extends classical mirror therapy into an **immersive 3-D scene** *— with real-time hand tracking and* **objective kinematic data**.



KEY DIFFERENCES FROM CLASSICAL MT

Built on classical MT, freed from the box

Keeps the multisensory **neuroplastic engine** of MT — replaces the bulky 2-D mirror with an immersive 3-D scene, real-time hand capture, and automatic kinematic logging.

- Static 2-D reflection → **immersive 3-D scene** with stereoscopic depth.
- Passive mirror → **real-time motion capture** of the unaffected hand.
- No dose tracking → **kinematic data** captured per session.
- **Engaging, gamifiable** — deployable at clinic and home.

Lin et al. IEEE Access 2021;9:14725–14734 · **Hsu H-Y, Kuo L-C, Lin C-W, Su F-C, Yang T-H, Lin C-W.** Neurorehabil Neural Repair 2022;36(6):335–345.

Virtual Reality Mirror Therapy (VRMT) from NCKU

CLINICAL USE & PATIENT PROFILE

CLINICAL INDICATIONS

Who benefits clinically?

- Patients with **unilateral cerebral infarction or hemorrhage** who have mild to severe motor impairments in the **proximal and distal upper limbs**.
- Patients with **hemiparesis** who do not have significant cognitive or perceptual impairments.

Targets functional recovery of the affected upper limb in adult stroke rehabilitation.

PATIENT TYPES

- Includes **stroke** and **cerebral palsy** patients.

CLINICAL USERS

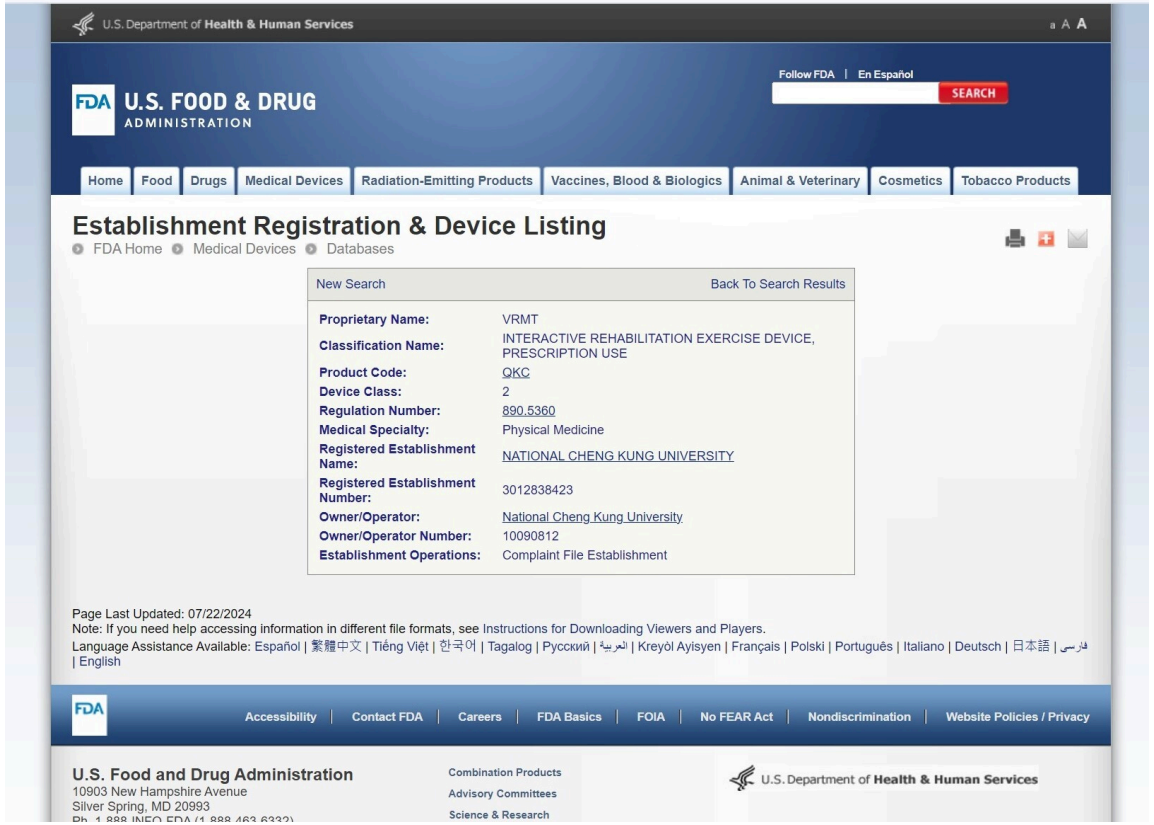
- **Occupational therapists (OTs)** and **physical therapists (PTs)**.
- Mirror therapy can be arranged by OTs/PTs in clinical rehabilitation settings.

Virtual Reality Mirror Therapy (VRMT) from NCKU

FDA REGULATORY STATUS & INVENTORS

FDA · ESTABLISHMENT REGISTRATION & DEVICE LISTING

CLASS II · 510(K) CLEARED



FDA REGISTRATION DETAILS

PROPRIETARY NAME	**VRMT**
CLASSIFICATION	Interactive Rehabilitation Exercise Device · Prescription Use
PRODUCT CODE	QKC
DEVICE CLASS	Class II · 510(k) Cleared
REGULATION NO.	21 CFR 890.5360
MEDICAL SPECIALTY	Physical Medicine
OWNER / ESTABLISHMENT	National Cheng Kung University

Source: U.S. FDA Establishment Registration & Device Listing · Updated 2024-07-22
NCKU news release: web.ncku.edu.tw/p/406-1000-270857

INVENTORS

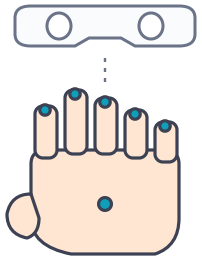
Prof. Hsiu-Yun Hsu
Department of Occupational Therapy, NCKU

Prof. Che-Wei Lin
Department of BioMedical Engineering, NCKU

What VRMT Does

An immersive **virtual-reality alternative** *to the classical mirror box — the unaffected hand is captured, mirrored, and rendered as the affected hand in a real-time 3-D scene,* **driving motor recovery for upper-limb hemiparesis after stroke**.

01 · CAPTURE

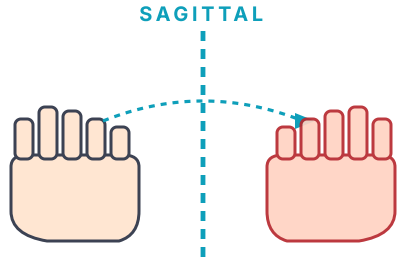


unaffected hand · 24 joints

(P_x, P_h, P_o)

real-time stream

02 · MIRROR

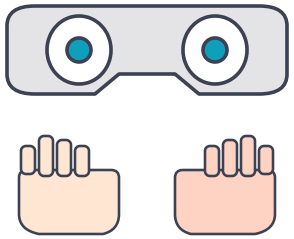


x ⟶ −x · affected hand

$(-P_x, P_h, P_o)$

mirrored coords

03 · RENDER



immersive 3-D scene · embodied

WHY IT WORKS

Multisensory neuroplasticity

Visual + bilateral proprioceptive integration drives motor learning.

Giroux et al. PLoS One 2018 · Maier et al. Front Syst Neurosci 2019

Bilateral cortical priming

Mirror feedback rebalances motor-cortex activity between hemispheres.

Rossiter et al. Neurorehabil Neural Repair 2015 · Stinear et al. Stroke 2014

Embodied limb ownership

Immersive 3-D scene strengthens sense of the affected limb.

Matamala-Gomez et al. Front Hum Neurosci 2019 · Hoermann et al. PLoS One 2012

CHAPTER 02

The VRMT Family

How the pipeline runs — capture, mirror, render — and the six feedback variants built on top.

SLIDES 13 — 24

02

Hand Motion Capture

THREE GENERATIONS OF VRMT INPUT

Capturing the unaffected hand drives the entire VRMT pipeline. Across three hardware generations the NCKU program has traded **dedicated accuracy** → **zero hardware** → **integrated immersion** *— each iteration shrinking the setup burden.*

GENERATION 1 FIRST VRMT PROTOTYPE

Leap Motion Controller

Desk-mounted infrared optical sensor — 2 IR cameras + 3 LEDs reconstruct hand structure via stereo vision.

0.01 mm precision · 200 fps · 2 hands × 10 fingers

+ **High kinematic fidelity** — dedicated tracking volume.
– Constrained interaction box; requires desk setup.

Lin et al. *IEEE Access 2021;9:14725* · **Hsu, Kuo, Lin et al.** *Neurorehabil Neural Repair 2022;36(6):335* · **Cho et al.** *Med Sci Monit 2018.*

GENERATION 2 MARKERLESS · LOW COST

Webcam + MediaPipe

Off-the-shelf RGB webcam feeding Google MediaPipe Hands — a deep-learning hand-landmark model.

21 keypoints/hand · 30 fps · ~95% rehab-task accuracy

+ **Zero dedicated hardware** — runs on any laptop or home PC.
– 2-D depth ambiguity; sensitive to lighting & viewing angle.

Camargos et al. *J NeuroEng Rehabil 2025* · **Amprimo et al.** *Biomed Signal Process Control 2024* · **Zhang et al.** *MediaPipe Hands, Google 2020.*

GENERATION 3 CURRENT · PRODUCTION

Oculus / Meta Quest HMD

Quest 2/3 built-in 4× IR camera array + on-device CV pipeline — no external sensor, no controller.

24 hand joints · ego-centric viewpoint · stereoscopic 3-D scene

+ **Tracking + immersive scene in one device** — truly hands-free.
– Dropout at extreme arm angles; HMD comfort for long sessions.

Abdlkarim et al. *Sci Rep 2025 (Quest 2 validation)* · **Iruthayarajah et al.** *Sci Rep 2023 (low-cost VR MT)* · **Meta** *Hand Tracking SDK 2023.*

Mirroring

SAGITTAL-PLANE REFLECTION OF THE UNAFFECTED HAND

The system reflects the captured unaffected-hand pose across the user's midline (sagittal plane), then renders it as the affected hand in an ego-centric scene — sustaining the multisensory illusion that re-engages the lesioned motor cortex.

WHY IT WORKS · NEURAL RATIONALE

Mirror Visual Feedback (MVF)

Congruent visual + proprioceptive input creates the illusion that the paretic limb is moving normally — bypassing **learned non-use** and re-engaging the lesioned hemisphere.

- Stimulates the lesioned **primary motor cortex (M1)**, premotor cortex, and parietal lobe.
- Reduces pathological **inter-hemispheric inhibition imbalance**.
- Drives **activity-dependent neuroplasticity** in the affected sensorimotor network.

Ramachandran & Rogers-Ramachandran *Proc R Soc B 1996* · **Thieme** *Cochrane 2018* · **Rossiter** *Neurorehabil Neural Repair 2015.*

HOW WE BUILT IT · NCKU IMPLEMENTATION

Meta Quest 2 + Oculus Interaction SDK

24 hand landmarks are tracked in real time without a controller. An **OVRCustomHandPrefab** renders the mirrored affected hand inside the user's ego-centric VR scene.

- **Visual deception design** — the mirrored hand is rendered in a realistic skin tone to maximise *sense of ownership*; the healthy hand is rendered in monotone gray to suppress visual interference.
- **VRMT-FOA extension** — real-time eye-tracking + dual-task (arithmetic / card-sort) locks attention on the paretic workspace and boosts motor-cortex *ERD*.

Lin et al. *IEEE Access 2021;9:14725* · **Hsu, Kuo, Lin et al.** *Neurorehabil Neural Repair 2022;36(6):335.*

POSITION TRANSFORM

$\texttt{MirrorPos} = (-P_x, P_y, P_z)$

Flip the X-coordinate across the sagittal plane; keep Y and Z.

ROTATION TRANSFORM

$\texttt{MirrorRot} = (R_x{+}180°, -R_y, R_z)$

Symmetric Euler-angle transform on the three rotation axes.

RENDER ENGINE

`Quaternion.Euler(…)`

Unity converts Euler angles to quaternion before driving the 3-D hand model.

Rendering

UNITY-BASED VRMT FAMILY

Each Unity-based system renders the mirrored hand into a distinct virtual scene tuned to its therapeutic goal. The pipeline math is shared (see s12) — only the scene assets and hand-model fidelity differ. (SM-ARMT's mobile AR stack → next slide.)

TO-VRMT

Oculus Rift + Unity

`Unity + C# · Oculus Rift @ 90 Hz`

Basic stick-and-ball **skeleton hands**; ADL objects rendered on a virtual table for task-oriented manipulation.

- Hand model: white rods + blue joint balls.
- Scene: **pens, playing cards, coins + containers**.
- Mirror plane: X-axis inversion along Y-axis.

Fauzan & Lin *Master's Thesis, NCKU 2020.*

MD-VRMT

Quest 2 + XR Toolkit

`Unity + XR Interaction Toolkit · Quest 2`

Audio-driven scene — a **Mirror Drum** with falling rhythm blocks; assets modeled in Blender, exported to Unity.

- Scene: Mirror Drum + falling rhythm blocks + drumsticks.
- Pipeline: **Blender → Unity** export.
- Dynamic visual score feedback per drum hit.

Chen Y-W, Hsu H-Y, Lin C-W. *WTMH Lab Poster.*

RH-VRMT

Quest 2 + Oculus SDK 50

`Unity 2021.3.8f1 · Oculus Interaction SDK 50.0`

OVRCustomHandPrefab in a virtual living-room scene; visual deception via skin-tone asymmetry.

- Hand: `SyntheticHand` smoothing + `HandVisual` mesh; dynamic palm scaling.
- Color: affected = realistic skin, healthy = monotone gray.
- **No virtual robot** in VR — TIGER acts physically via Arduino.

Liu & Lin *Master's Thesis, NCKU 2024.*

SHARED / UNITY FAMILY PIPELINE

TO · MD · RH all use the same C# transform script before mesh rendering: `MirrorPos = (−Pₖ, Pⱼ, Pᵢ)` + `Quaternion.Euler(Rₖ+180°, −Rⱼ, Rᵢ)` — only the scene assets and hand-model fidelity differ across systems.

Rendering · Mobile AR

INSIDE THE IOS-NATIVE PIPELINE

SM-ARMT departs from Unity entirely. *The pipeline runs natively on iOS, extracts the user's hand as a 2-D contour, and overlays it onto the real-world camera feed — no 3-D mesh, no HMD, no PC.*

ARCHITECTURE

SwiftUI + Metal + ARKit 3

A custom iOS app processes the rear-camera stream entirely on-device. Different stack for a different use case — no Unity, no Oculus SDK.

- **SwiftUI** — declarative UI shell for the iOS app.
- **Metal** — Apple's GPU API powers the overlay shader + Cardboard barrel-distortion.
- **ARHeadsetKit** — Swift package for Cardboard dual-view rendering.
- **ARKit 3 Person Segmentation API** — system-level human segmentation (replaces a custom U-Net prototype for ~3× speedup).

VISUAL DESIGN

2-D Hand Contour, No 3-D Mesh

Instead of rendering a 3-D hand model, the system extracts the unaffected hand as a 2-D contour and pastes it — mirrored — directly onto the real-world camera background.

- **Motion-only mirroring** — hand pixels only; no body, no virtual environment.
- Avoids the **uncanny valley** of full-body 3-D rendering.
- **Cardboard support** — dual-view + barrel distortion for stereoscopic parallax.
- Patient's own real-world background = familiar, low-distraction context.

PERFORMANCE & PIPELINE

60 FPS overlay at **10–20 ms** inference · **iPhone 11+** (iOS 15+ / A12 Bionic+) · **zero PC/HMD** — most accessible family delivery.

ML PIVOT Custom **U-Net + MobileNetV2** (dice 0.124, 35–40 ms, 25–28 FPS) → **ARKit 3 Person Segmentation** (10–20 ms, 60 FPS) · Cardboard via Metal shader · CPU:GPU 2:8 + shared frame buffer

Huang P-Y & Lin C-W. Development of iOS-based Augmented Reality Mirror Therapy Software for Upper Limb Rehabilitation in Stroke-Induced Hemiparesis. Master's Thesis, National Cheng Kung University, 2023.

Research Journey

EIGHT YEARS OF VRMT/ARMT AT NCKU WTMH LAB

2018 — **First VRMT prototype**
Lin C-W lab · Leap Motion + Oculus Rift, baseline mirror-feedback engine.

2020 — **TO-VRMT — task-oriented variant**
Fauzan L & Lin C-W · NCKU master's thesis · 2-arm pilot RCT (n=9 + n=9 healthy adults); MMDT Turning between-group sig.

2021 — **IEEE Access publication**
Lin C-W, Kuo L-C, Hsu H-Y et al. · 3-arm RCT, n=54 chronic stroke (CONSORT in slide 19).

2022 — **NNR clinical validation**
Hsu, Kuo, Lin et al. Neurorehabil Neural Repair 36(6):335 · sensorimotor recovery confirmed.

2023 — **SM-ARMT — mobile AR on iOS**
Huang P-Y & Lin C-W · thesis · fNIRS-verified upper-limb gains vs traditional MT.

2024 — **RH-VRMT — robotic TIGER + soft glove**
Liu T-H & Lin C-W · thesis · synchronous robot mode + EEG coherence gains.

2025 — **VRMT-FOA + FaceMirror**
Huang A-T, Hsu H-Y, Lin C-W gaze-guided VRMT-FOA (+98% gaze) · FaceMirror facial-palsy feasibility.

2026 — **Taiwan TFDA Class I — target (you are here)**
Regulatory clearance milestone · consumer-mobile deployment path.

2027+ — **US FDA 510(k) → AI Upgrade (2028+)**
Planned roadmap: clinical scaling + diagnostic-AI augmentation.

VRMT & ARMT Family

SIX FEEDBACK MODALITIES · ONE MIRROR-THERAPY ENGINE

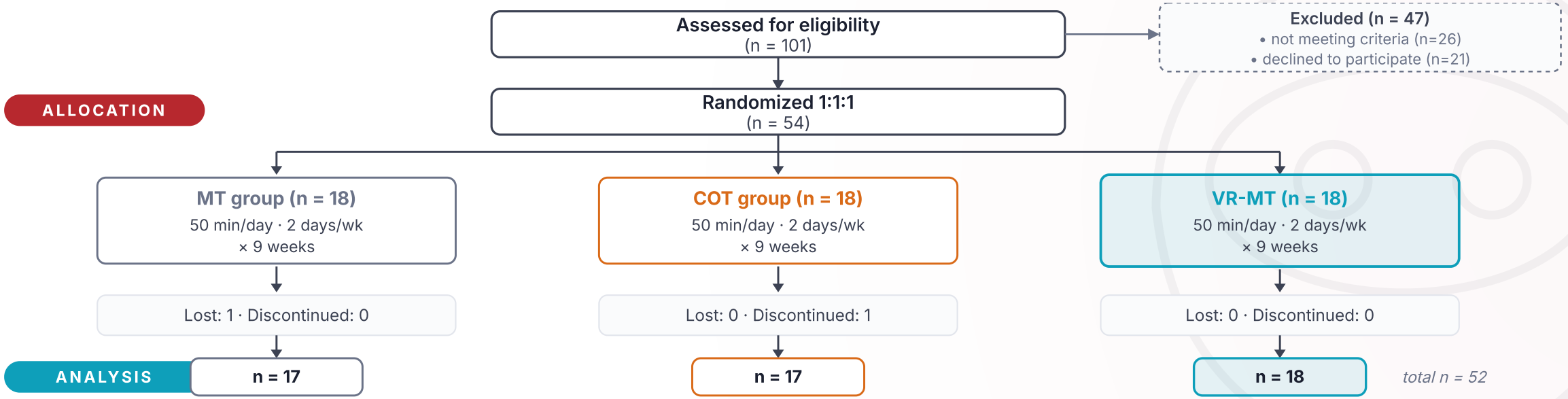
We extend the baseline visual mirror illusion with progressively richer feedback — task-oriented cues, music + vibration, active robotic movement, gaze-guided attention — and finally drop the headset entirely for mobile AR telerehab.

SERIES	SM-VRMT Simple	TO-VRMT Task-Oriented	MD-VRMT Music-Driven	RH-VRMT Robotic-Hand	SM-ARMT Mobile AR	VRMT-FOA Gaze-Guided
SCENE						
FEEDBACK	Visual	Visual + Sensory	Visual + Audio + Haptic	Visual + Movement	Visual iOS AR	Visual + Gaze + Audio
HARDWARE	Quest 2 + Leap Motion	Oculus Rift + Leap Motion	Quest 2 + Controller + OpenBCI	Quest 2 + TIGER + soft glove	iPhone (iOS) + Cardboard	Quest Pro + OpenBCI Cyton
DEEP DIVE	→ SLIDE 19	→ SLIDE 20	→ SLIDE 21	→ SLIDE 22	→ SLIDE 23	→ SLIDE 24

SM-VRMT — Simple Virtual Reality Mirror Therapy

BASELINE · VISUAL FEEDBACK ONLY

The foundational system — replaces the classical mirror box with a fully immersive 3-D VR scene. **No extra feedback modality beyond visual MVF.**



HARDWARE & METHOD

Meta Quest 2 + Leap Motion (sub-mm, 200 fps). Visual MVF only — no audio, haptic, or robotic assistance.

PILOT RESULTS

Stroke patients showed **significant** gains in **FMA-UE** + **BBT** ($p < .05$); SM-VRMT > traditional MT.

TO-VRMT — Task-Oriented Mirror Therapy

+ TASK-ORIENTED SENSORIMOTOR FEEDBACK

Replaces simple repetitive joint movements with **functional daily-life tasks** *— pen-grasping, card-flipping, coin placement — to combat boredom and sustain visual attention.*



HARDWARE & METHOD

Oculus Rift + Leap Motion

Functional task scenarios are placed in the affected-limb workspace; patients perform ADL-like manipulations while receiving mirror visual feedback.

- **Headset**: Oculus Rift.
- **Hand tracking**: Leap Motion controller.
- **Feedback**: visual MVF + **task-oriented sensorimotor cues** from interacting with virtual objects.

STUDY DESIGN & OUTCOMES

2-Arm Pilot RCT · Healthy Young Adults

Short-term efficacy trial vs traditional task-oriented mirror box, with 2-hour follow-up.

- **Cohort**: **n=9 VRMT vs n=9 Trad MT** (age ~22 yr).
- **Protocol**: 30-min single session; assessed at `T0` · `post` · `2-h follow-up`.
- **Comparator**: traditional task-oriented mirror box.
- **Outcomes**: `PHUA`, `MMDT` (turning + placing), `PPT` (assembly), `SWM`, Grasp Power.

PILOT RESULTS

+ MMDT Turning — the only **between-group significant** outcome: VRMT **80.8 → 66.6 s** vs Trad 86.0 → 71.6 s (**F=5.289, p=.036, η^2=.261**).
± Within-group sig: PHUA FP/FR_{peak} ($p \leq .002$) · MMDT both subtests (VRMT). **– n.s.**: PPT Assembly trend (p=.06), SWM, Grasp Power.

ML PIPELINE Leap Motion **100 Hz** + 4 Hz low-pass (order 3) + **LSTM** trajectory prediction · X-axis RMSE **0.0337**

Fauzan L & Lin C-W. Development of a Virtual Reality Rehabilitation System based on Task-Oriented Mirror Therapy and its Verification on Healthy Subjects. Master's Thesis, National Cheng Kung University, 2020.

MD-VRMT — Music-Driven Mirror Therapy

+ RHYTHMIC AUDIO + VIBRATION FEEDBACK

Adds synchronous rhythmic music and controller vibration to the visual mirror illusion — a tri-modal feedback loop (visual + audio + haptic) on a Mirror-Drum interface.



HARDWARE & METHOD

Quest 2 + Controllers + OpenBCI

Synchronous music plays as the patient strikes a virtual "Mirror Drum"; VR controllers provide haptic vibration; EEG monitors cortical activation.

- **Headset**: Meta Quest 2.
- **Peripherals**: VR Controllers (haptic vibration) + **OpenBCI Cyton** EEG cap.
- **Feedback**: visual + **audio** + **haptic**.

STUDY DESIGN & OUTCOMES

Stroke Parallel RCT + Healthy Crossover

Two distinct study designs — clinical efficacy on stroke + mechanism probing on healthy.

- **Stroke**: **n=20** chronic hemiplegic, **2-arm parallel 1:1** (Mirror Drum vs traditional MT; usual care given to both).
- **Stroke dose**: 50 min therapy + 20 min usual care, 2 days/wk × **9 weeks**.
- **Healthy**: **n=60**, **3-condition crossover 1:1:1** — Mirror Drum w/ music · w/o music · MT (10 min EEG + 10 min therapy, 1-wk washout).
- **Outcomes**: stroke — `FMA-UE` , `MAS` , `BBT` , `SWM` , `MAL` , `CGI` ; healthy — `PHUA` + EEG `ERD/Coherence` .

PILOT RESULTS

Stroke: FMA-UE total **p=.007**, shoulder **p=.001** vs MT. **Healthy EEG**: music+vibration ↑ Mu/Beta ERD — *qualitative only*.

MD PIPELINE **Demucs** source-separation · OpenBCI Cyton 1–50 Hz · **pwelch()** PSD + **mscohere()** coherence

Caveats: n=20 stroke · only shoulder p=.001 · no effect sizes/CIs · healthy EEG qualitative only

Chen Y-W, Hsu H-Y, Lin C-W. Rhythmic Music + Vibration Stimulation for VRMT. WTMH Lab Poster.

RH-VRMT — Robotic-Hand Mirror Therapy

+ ACTIVE MECHANICAL MOVEMENT (PROPRIOCEPTIVE)

Adds **active mechanical movement** *of the affected hand via a wrist exoskeleton + soft glove — visual MVF gains a* **proprioceptive partner**.



HARDWARE & METHOD

Quest 2 + TIGER Exoskeleton + Soft Glove

Two operating modes: **synchronous** — the healthy hand drives the robotic device on the affected hand; **passive** — the robot drives the affected hand independently.

- **Headset**: Meta Quest 2 + Leap Motion.
- **Robotics**: Arduino Uno + **TIGER** wrist exoskeleton + **soft ROBOT** glove.
- **EEG**: OpenBCI Cyton for cortical activation + coherence.

STUDY DESIGN & OUTCOMES

5-Condition Crossover vs Pure VRMT

2 robots (TIGER + soft ROBOT) × 2 modes (synchronous / passive) + baseline.

- **Cohort**: **n=30** right-handed healthy young adults.
- **Conditions**: `VRMT` · `Syn + soft ROBOT` · `Alt + passive soft` · `Syn + TIGER` · `Alt + passive TIGER`.
- **Protocol**: 5 × 20-min sessions, **1-week washouts**; randomized order.
- **Outcomes**: `PHUA`, `PPT`, `MMDT`, `2PD`, `SWM` + EEG `ERD/ERS · Coherence`.

PILOT RESULTS

Syn VRMT + TIGER vs VRMT: significantly stronger **Beta ERD** in right-lateralized frontal areas — right prefrontal **p=.019**, frontal **p=.03**, right frontal **p=.012**. **Frontal-occipital coherence** also gains (F3–O2 p=.036, F7–O2 p=.037) — objective evidence the robotic add-on engages broader cortical networks.

CAVEATS `healthy n=30 ceiling effect · OpenBCI Cyton+Daisy (consumer-grade EEG, not medical) · pneumatic-pump latency limits real-time sync · rest-vs-movement eye-closure confounds ERD attribution`

Liu T-H & Lin C-W. Hybrid System for Robot-Assisted VR Mirror Therapy Targeting Distal Joints of the Upper Limb. NCKU master's thesis, 2024.

SM-ARMT — Simple Augmented Reality Mirror Therapy

IOS-BASED · MOBILE TELEREHABILITATION

Drops the VR headset entirely — a smartphone's rear camera + ML body-segmentation network mirrors the hand onto a real-world background for accessible at-home telerehab. Equipment economy: **VR-MT stack $1k–15k → smartphone $300–800 + Cardboard $3–20.**

HARDWARE & METHOD

iPhone (iOS) + Cardboard

A custom iOS app processes the rear-camera stream through a human-body semantic-segmentation neural net at **60 fps**; only the hand motion is mirrored, avoiding the uncanny-valley effect of full-body rendering.

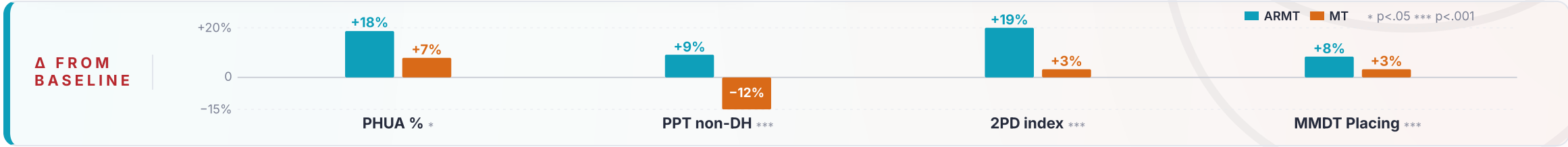
- **Device**: iOS smartphone (iPhone 11 tested) + optional Cardboard headset for 3-D split-screen view.
- **Method**: motion-only mirroring on real-world background.
- No expensive VR/PC setup — pure mobile.

STUDY DESIGN & FINDINGS

Crossover RCT vs Traditional MT

Within-subject design with both behavioral and cortical-perfusion measurement.

- **Cohort**: **n=30** young healthy, 30-min ARMT + 30-min MT, 1-week washout.
- **Outcomes**: PHUA, PPT, MMDT, 2PD, SWM + fNIRS prefrontal.
- **fNIRS**: bilateral prefrontal synchrony ρ>0.9 in *both* arms; **no group difference** — equivalent neural signature.
- **Behavior**: ARMT > MT on **9 of 12** outcomes (p<.05); see chart ↓



MOBILE PIPELINE **ARKit 3 Person Segmentation** · 10–20 ms → **60 FPS** · **iOS 15+ / A12 Bionic+** floor

Bars show **Δ% from baseline**; the **−12% on PPT non-DH** is the traditional-MT control arm *regressing* — mobile ARMT preserves while MT actively interferes on the dominant–non-dominant transfer task. **Huang P-Y & Lin C-W.** iOS-based AR Mirror Therapy Software for Upper-Limb Rehabilitation in Stroke-Induced Hemiparesis. NCKU master's thesis, 2023.

VRMT-FOA — Gaze-Guided Mirror Therapy

FOCUSED OBJECTIVE ATTENTION · NEURAL EVIDENCE

In standard VRMT, patients drift into an autopilot state — eyes wander, sensory neglect returns. VRMT-FOA locks attention on the paretic workspace with a decide-then-act dual task, and verifies it with eye-tracking + EEG.

GAP IN REGULAR VRMT

Autopilot & Sensory Neglect

Patients passively receive the mirror illusion; after a few minutes they perform repetitive motor tasks without active visual or cognitive engagement — rehab efficacy plateaus.

- No objective measure of **where the patient is looking**.
- Goal-directed attention is what drives **synaptic modification & activity-dependent plasticity**.

FOA DESIGN

"Decide-then-Act" Dual Task

A cognitive challenge is presented in the affected-limb workspace; the patient must **solve, then move** — forcing sustained focus on the paretic side before and during execution.

- **Move Forward** + Math · **Pull Out** + Color Matching · **Circle** + Reading
- **Put Coins** + Color Matching · **Shoot Ball** + Counting · **Flip Cards** + Visual Matching

HARDWARE DELTA

Quest Pro + OpenBCI Cyton

Step up from Quest 2 to **Meta Quest Pro** for built-in eye-tracking, paired with continuous EEG via **OpenBCI Cyton**.

- **Eye-tracking** — real-time gaze trajectories on the patient's view.
- **EEG** — Mu (8–13 Hz) + Beta (13–30 Hz); thesis-explicit ROIs `C3 / C4` motor cortex, plus prefrontal + parietal.

PILOT RESULTS

AFFECTED-LIMB GAZE FIXATION · PER SCENE

+77% → +98%

VRMT 10.8 s *(gaze ratio 36%)* → **VRMT-FOA 19.2 s · 64%** → VRMT-FOA-AF 21.4 s · 71%.

CORTICAL ACTIVATION (MU / BETA ERD)

Strongest, most stable ERD

Peak desynchronisation across **motor (C3/C4) · prefrontal · parietal** cortices — indicating sustained motor-preparation + attention.

FOA PROTOCOL **1:1:1 crossover** · 3 randomized orders (VRMT→FOA→FOA-AF / FOA-AF→VRMT→FOA / FOA→FOA-AF→VRMT) · per condition: 10 min EEG + 10 min eye-tracking · n=20 young healthy

Huang A-T, Hsu H-Y, Lin C-W. VRMT-FOA: Gaze-Guided VR Mirror Therapy Enhancing Cortical Activation + Network Coherence. NCKU pilot crossover, n=20 young healthy adults.

CHAPTER 03

Evidence & Direction

Clinical trials, citation impact, biomarker signatures — and where the AR thrust is heading.

SLIDES 26 — 34

Clinical Evidence · Stroke RCTs

TWO OF SIX SYSTEMS HAVE FULL HEMIPLEGIA TRIALS

Of the six family systems, only SM-VRMT and MD-VRMT have full chronic-stroke RCTs. The other four (TO / RH / ARMT / VRMT-FOA) were validated on healthy adults first — stroke validation is the next translational step.

SM-VRMT · BASELINE

Three-Arm RCT, 9–12 Weeks

First validation of digital MVF — immersive VR replaces the physical mirror box for chronic-stroke hemiplegia patients.

- **Population**: **n=54** chronic stroke hemiplegia.
- **Design**: 3-arm RCT — VRMT vs conventional MT vs conventional OT.
- **Outcomes**: `FMA-UE`, `Box and Block Test`.
- **Result**: **Significant** gains in wrist motion + coordination; FMA-UE change exceeds controls.

MD-VRMT · MULTISENSORY

Multisensory Stroke RCT

9-week multisensory intervention — Mirror Drum + synchronous music + vibration vs traditional MT (both arms also receive usual care).

- **Population**: **n=20** chronic hemiplegic stroke.
- **Design**: **2-arm parallel RCT**, 50 min therapy + 20 min usual care, 2 days/wk × 9 weeks.
- **Outcomes**: `FMA-UE`, `MAS`, `BBT`, `SWM`, `MAL`, `CGI`.
- **Result**: FMA-UE total **p=.007**; shoulder subscale **p=.001** (elbow/forearm not reported).

EVIDENCE GAPS

Drop-out, blinding, long-term follow-up not reported across the family. **CI / Cohen's d** not provided — only p-values and percentage changes. Stroke RCTs for **TO / RH / ARMT / VRMT-FOA** remain pending translation work.

Lin C-W et al. IEEE Access 2021;9:14725 · **Hsu, Kuo, Lin et al.** Neurorehabil Neural Repair 2022;36(6):335 · **Chen Y-W, Hsu H-Y, Lin C-W.** WTMH Lab Poster.

Citation Reception

HSU 2022 NNR IN SUBSEQUENT SYSTEMATIC REVIEWS

Hsu, Kuo, Lin et al. NNR 2022** — the 3-arm RCT (n=54, MT / COT / VR-MT) anchoring SM-VRMT — has been re-analyzed in **5+ systematic reviews (2023–2025)**. The pooled verdict: **consistently positive, methodologically "moderate quality."

INCLUDED IN · 5 RECENT REVIEWS

2023–2025 Systematic Reviews + Meta-Analyses

- **Tian et al. 2023** · *Front Neurol* — VR-MT specific meta (most direct).
- **JMIR 2025** e73142 — Combined immersive + non-immersive VR + MT.
- **Saragih et al. 2025** · *Clin Rehabil* — 18 RCTs, n=633 stroke.
- **Hsieh et al. 2025** · *PM&R* — subacute MT meta-analysis.
- **Hao 2024** · *Health Sci Rep* — VR stroke rehab umbrella review.

POOLED EFFECT SIZES

Direction: significant + positive

Across meta-analyses including Hsu 2022, VR-MT outperforms traditional MT on motor + dexterity outcomes.

- **Tian 2023**: FMA-UE `SMD 0.81` (*large*) · BBT `SMD 0.48` · MFT `SMD 0.72`.
- **JMIR 2025**: FMA-UE `MD 3.50` (95% CI 1.47–5.53; p=.0007) · BBT `MD 1.09` (p=.03).
- **Hsu's own**: FMA-UE +3.8 (9wk) / +4.2 (12wk) · BBT +2.9 / +3.1 · hand subscale **p=.01** vs MT.

STRENGTHS VS LIMITS

+ Strengths: rare **3-arm design** (MT · COT · VR-MT), **12-wk follow-up**, sig. hand-subscale gain, consistently positive across reviews.
– Limits: small **n=54**, no VR blinding, pooled FMA-UE *below MCID*, no long-term follow-up.

Tian Front Neurol 2023 · **JMIR** 2025;27:e73142 · **Saragih** Clin Rehabil 2025 · **Hsieh** PM&R 2025 · **Hao** Health Sci Rep 2024. *(Original: Hsu et al. NNR 2022;36(6):335.)*

Objective Biomarkers

EEG · FNIRS · GAZE TRACKING ACROSS THE FAMILY

Beyond clinical scales, the program quantifies **mirror-therapy mechanisms** *with cortical (EEG), hemodynamic (fNIRS), and oculomotor (gaze) biomarkers — objective evidence for* **how the therapy works**.



MD-VRMT — Multisensory EEG

n=60 healthy **3-condition crossover** (Mirror Drum w/ music · w/o music · MT). Adding music + vibration visibly increased **Mu / Beta ERD** in the motor cortex — *qualitative* observation; poster does not report exact p-values.

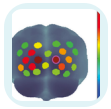
Marker: *EEG ERD & Coherence* · **Chen Y-W, Hsu H-Y, Lin C-W.** *WTMH Poster.*



RH-VRMT — Robot + EEG Coherence

Healthy crossover (5 conditions). **Syn VRMT + TIGER vs VRMT**: stronger **Beta ERD** in right-lateralized frontal areas (right prefrontal p=.019 · frontal p=.03 · right frontal p=.012); frontal–occipital coherence gains (F3–O2 p=.036, F7–O2 p=.037).

Marker: *EEG Beta ERD + Frontal–Occipital Coherence* · **Liu & Lin** *2024.*



SM-ARMT — fNIRS Hemodynamics

n=30 healthy crossover. **Prefrontal** bilateral synchrony equally strong in both arms (ARMT ρ=.961 · MT ρ=.969). **Motor cortex** bilateral synchrony notably *lower* in ARMT than MT (**ρ=.344 vs .782**) — mobile AR engages frontal attention well but recruits less bilateral motor cooperation. ARMT still wins behaviorally on 9 of 12 clinical outcomes (slide 32).

Marker: *fNIRS prefrontal perfusion* · **Huang & Lin** *2023.*



VRMT-FOA — Gaze + Strongest ERD

Affected-side gaze ratio **36% → 64% → 71%** (VRMT → VRMT-FOA → +Audio); gaze-duration **+77%** / **+98%**. **Strongest, most stable Mu / Beta ERD** across motor, prefrontal, parietal cortices; VRMT-FOA-AF also showed the *most persistent inter-regional coherence* (qualitative).

Marker: *Eye-tracking + EEG ERD/Coherence* · **Huang A-T, Hsu H-Y, Lin C-W.**

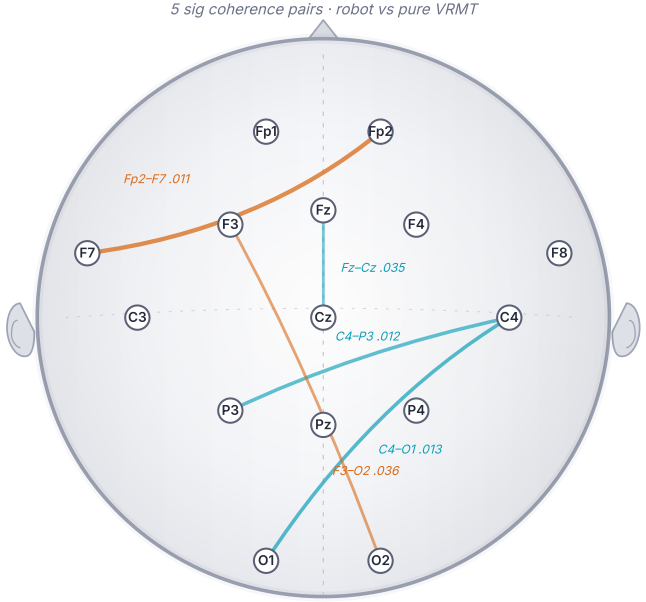
Cortical Network Engagement

RH-VRMT EEG COHERENCE · ROBOT AUGMENTS INTER-REGIONAL CONNECTIVITY

Robot-augmented conditions enhance **EEG coherence** *not only during active hand movement, but also at* **rest** *— suggesting neural carryover beyond active engagement. Strongest single effect:* ***Fz–O1 Mu p<.001*** *(rest phase, Syn TIGER vs Alt TIGER).*

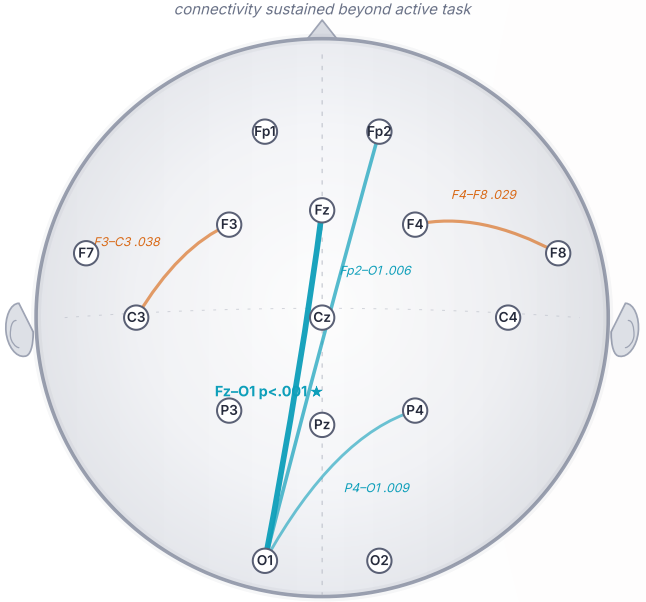
HAND-MOVEMENT PHASE

5 sig coherence pairs · robot vs pure VRMT



REST PHASE

connectivity sustained beyond active task



REPRESENTATIVE PAIRS

Hand-movement phase · robot vs pure VRMT

- **Mu band** (8–12 Hz)
- **Beta band** (13–30 Hz)

9 findings · 30+ sig pairs across 4 robot conditions vs VRMT

Strongest: `Fz–O1 p<.001` (rest, Syn TIGER vs Alt TIGER)

PHASE-SPECIFIC THEMES

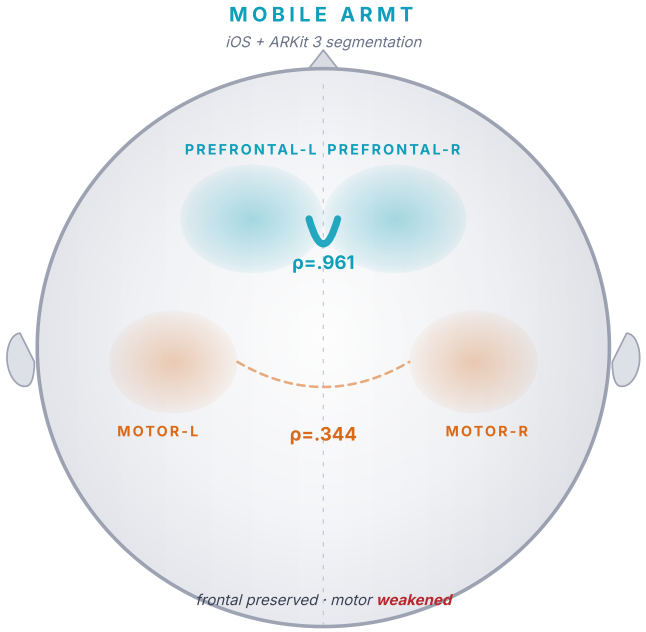
- **Movement**: motor → occipital/parietal (Mu) + bilateral frontal recruitment (Beta) + central-vertical axis (Mu).
- **Rest**: *right-hemisphere vertical* Fz–O1 **p<.001** dominates; left central-frontal (Beta) sustained; occipital suppression also present.
- **Interpretation**: connectivity *does not collapse* at rest — robot effect carries over, suggesting neural plasticity beyond task period.

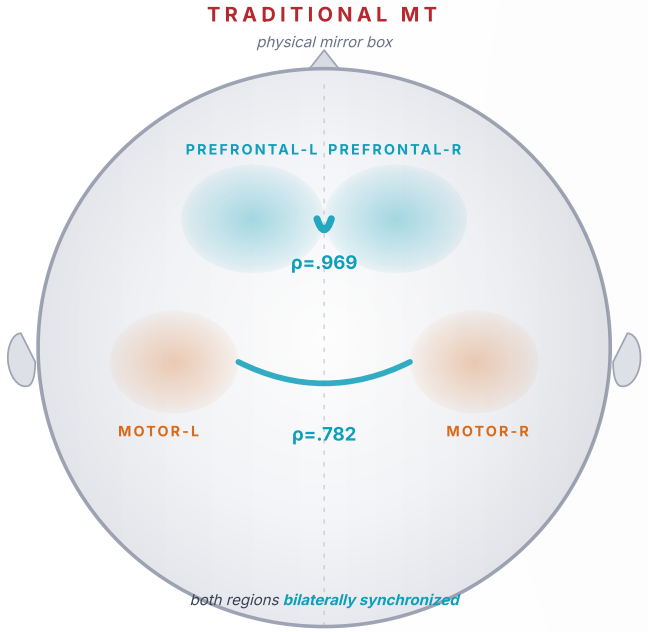
Liu T-H & Lin C-W. RH-VRMT EEG coherence (mscohere, Hand Movement + Rest phases). NCKU master's thesis, 2024. Cross-system context: Huang P-Y 2023, Huang A-T 2025, Chen Y-W WTMH Poster.

Bilateral Hemispheric Synchrony

SM-ARMT FNIRS · ARMT VS TRADITIONAL MT · PREFRONTAL SYMMETRIC, MOTOR ASYMMETRIC

In healthy subjects, **prefrontal L–R synchrony stays high in both arms**; *but* **motor-cortex bilateral synchrony drops sharply in ARMT** — *mobile AR engages frontal attention without recruiting equivalent bilateral motor cooperation. Behavioral superiority persists (slide 32).*





BILATERAL L–R SYNCHRONY (P)

Pearson correlation between L & R hemispheric HbO — per region

- **ρ > 0.7** : strong bilateral coupling
- **ρ < 0.5** : weak / asymmetric (dashed)

NIRScout: 16 detectors + 8 sources · 3 cm spacing

Tracked over **prefrontal cortex** + **motor cortex** (4 regional quadrants)

KEY FINDINGS

- **Prefrontal** bilateral synchrony stays robust (ρ ≈ .97) regardless of condition — frontal attention recruited equally.
- **Motor cortex** bilateral synchrony drops to **ρ=.344** in ARMT vs **.782** in MT — mobile AR fails to engage bilateral motor cooperation.
- **GLM activation**: no significant group difference in any region (all p>.05) — "similar excitation patterns" per author. Asymmetry shows in *connectivity*, not activation magnitude.

Huang P-Y & Lin C-W. iOS-based AR Mirror Therapy. NCKU master's thesis, 2023 (Table 8.2 GLM p-values; Figure 23 fNIRS layout). Behavioral comparison see slide 23 + slide 32.

From the Data

VRMT-FOA · ATTENTION-LOCKING IMPACT

AFFECTED-LIMB GAZE FIXATION · PER SCENE · VS VRMT BASELINE 10.8 S (GAZE RATIO 36%)

+77%

VRMT-FOA

*10.8 s → 19.2 s · gaze 36% → **64%***

+98%

VRMT-FOA + AUDIO FEEDBACK

*10.8 s → 21.4 s · gaze 36% → **71%***

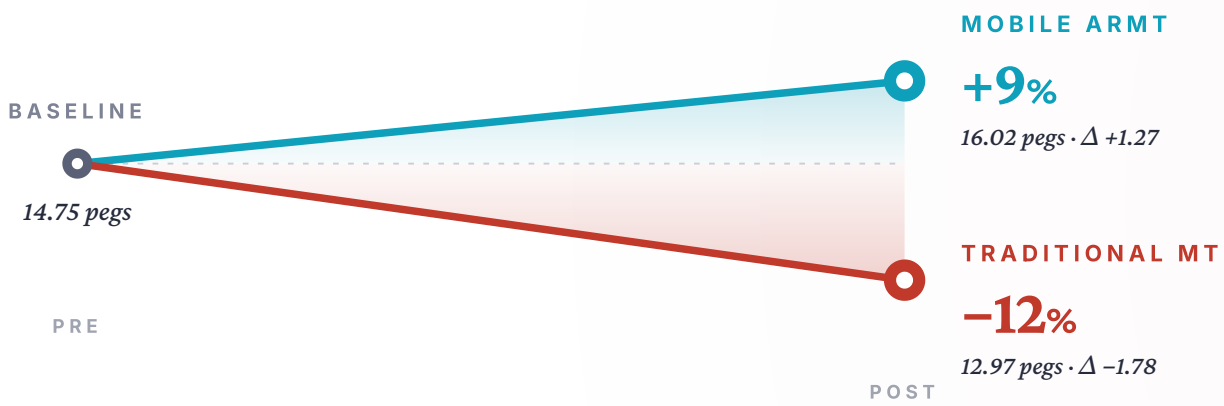
The **"decide-then-act"** *dual-task design* ***nearly doubles*** *the time patients spend looking at their paretic workspace — objectively measured via Quest Pro eye-tracking. Affected-side gaze ratio rises in 3 steps:* **36% → 64% → 71%.**

Huang A-T, Hsu H-Y, Lin C-W. VRMT-FOA: Gaze-Guided VR Mirror Therapy. Pilot crossover RCT, n=20 healthy adults.

From the Data

SM-ARMT · MOBILE, YET STRONGER

PURDUE PEGBOARD · NON-DOMINANT HAND · HEALTHY CROSSOVER N=30



Two treatments, same starting hand. Traditional MT pulls the trajectory **down**; *mobile ARMT pushes it* **up** *(p=.001). Across all measures, ARMT significantly outperforms MT on* **9 of 12 clinical outcomes** *(p<.05) — see the per-outcome breakdown on slide 23.*

Huang P-Y & Lin C-W. SM-ARMT vs Traditional MT, within-subject crossover, n=30 healthy adults. NCKU master's thesis, 2023 (Table 7.1 / 7.2).

AR Direction · From Phone to Future

MIRROR THERAPY BEYOND THE HEADSET

*AR democratizes the program — **unlimited home training** for less than one PT session; no $1–15K headset, no pre-set safety zone, real-world context preserved. The lab's AR thrust runs on consumer iPhones — **no AR-glasses pivot planned**.*

WHY AR MOTIVATION

Accessibility + Ownership

VR costs **$1–15K** + needs a safety zone — clinic-only. Consumer AR runs on phones patients already own.

+ **Unlimited home training** for less than one PT session.
+ **Own skin + lighting** — higher ownership than a generic VR avatar.

SM-ARMT UPPER LIMB

iOS + Core ML Segmentation

2-D contour overlay at **60 FPS** via Core ML on iPhone 11+. Crossover **n=30** healthy.

+ **PHUA FRpeak p=.003**, Percentage **p<.001**.
+ **PPT Assembly p<.001**, **2PD p<.001**, **MMDT p<.001**.
+ fNIRS prefrontal correlation **>0.9** vs traditional MT.

FACEMIRROR FACIAL PALSY

ARKit + Face Mesh

Motion vectors from healthy side → 3-D mesh of affected side. Preserves real facial asymmetries (moles / scars stay correct).

+ **SFGS 32 → 52 (+62%)** in 2 wks · FDI **+5%** (Social **+3.5**).
+ 30 min/day × 14 days · **n=1 pre-post feasibility**.

FACEMIRROR VS **MEPP** *(2D flip — moles wrong side)* · **Specular Face** *(uncanny valley)* · **FaraPy** *(no MNS, non-FP-trained AI)*

REGULATORY ROADMAP

2026 — **TAIWAN TFDA**
Class I biofeedback · offline-first

2027 — **US FDA**
510(k) · MindMaze predicate · N=30–50

2028+ — **AI UPGRADE**
Class II · insurance + pharma B2B

consumer mobile only

No AR-glasses, Vision Pro, or Hololens transition planned — *software / AI bets, not headset bets.*

Program Positioning

WEARABLE TECHNOLOGY & MOBILE HEALTHCARE LAB · 2026

*An **FDA-certified cornerstone** combined with EEG, fNIRS and eye-tracking objective quantitative data and full RCTs — building a comprehensive neurorehabilitation evidence system from high-end clinical equipment to ubiquitous home telerehabilitation.*

WTMH LAB · NCKU BME